ON A COHOMOLOGICAL DESCRIPTION OF CONGRUENCES BETWEEN MEROMORPHIC MODULAR FORMS AND ELLIPTIC CURVES

by

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1. Introduction

The aim of this paper is to give a cohomological interpretation of certain congruences between Fourier coefficients of modular forms and Frobenius eigenvalues arising from the geometry of modular (and Shimura) curves that arose in a recent work of Zhang [**Zha25**]. More precisely, we study congruences for overconvergent and meromorphic modular forms through the de Rham, rigid, and crystalline cohomology of natural coefficient systems attached to the universal abelian variety.

Let $p > 3$ be a prime and $N > 3$ an integer coprime to p. We work either with the modular curve $X_1(N)$ or, more generally, with a Shimura curve of quaternionic type, both defined over $\mathbb{Z}_p$. Denote by ω the line bundle of invariant differentials and by $(\mathcal{H}, \nabla)$ the rank-two vector bundle given by relative de Rham cohomology of the universal object, equipped with its Gauss–Manin connection. For an integer $k \geq 0$, we consider the symmetric power $\mathcal{H}_k = \mathrm{Sym}^k\mathcal{H}$, which carries a natural filtration and a logarithmic connection satisfying Griffiths transversality.

A key point is that the de Rham cohomology of $(\mathcal{H}_k, \nabla)$ on suitable open curves admits an explicit description in terms of modular forms modulo iterates of the θ-operator. This provides a bridge between cohomology classes and q-expansions, allowing arithmetic information to be extracted from Frobenius actions on cohomology. In particular, by comparing de Rham, rigid, and logarithmic crystalline realizations, one obtains canonical Frobenius-stable lattices whose images under the q-expansion map yield congruences for Fourier coefficients.

The guiding philosophy is that congruences for modular forms should be understood as shadows of Frobenius structures on geometric cohomology groups. From this perspective, classical results of Coleman on overconvergent modular forms and the slope theory of the U_p-operator arise naturally from the geometry of the supersingular locus, while new congruences for meromorphic modular forms can be interpreted via the Frobenius action on the de Rham cohomology of the poles.

1.1. An important note on a recent result. — A recent preprint of Allen, Long and Saad [**ALS25**] came out in November 2025 during the writing of this article. The authors analyze the same congruences between meromorphic modular forms and elliptic curves and give a proof of many of Zhang conjectures. Their methods extend the classical work of Scholl [**Sch85**] and a more recent work of Kazalicki-Scholl [**KS13**] to the case of meromorphic modular forms. The works are similar under many aspects, in particular the initial cohomological setting and the Gysin sequence is used in fundamentally the same way. Our approach differs from theirs primarily in the use of log-crystalline cohomology to identify a Frobenius stable lattice and the use of an explicit Frobenius lift enlarging the set of poles to the set of supersingular points. This allows us to avoid computing with local expansions at the cusp infinity, enlarge our framework to the Shimura setting and derive relation on Fourier coefficients only at the very end. On the other hand, their precise work takes care of the order of the poles of the meromorphic modular forms allowing them to obtain very sharp congruences on the cohomological subgroup.

After a very kind and fruitful exchange of emails with the authors, where they have looked at a first draft of this work and at the techniques involved, they allowed us to publish our version.

2. Geometric setting

Let $p > 3$ be a prime, let $N > 3$ be an integer coprime with p. Consider one of the following setting

- $X = X_1(N)$ modular curve of level $\Gamma_1(N)$ defined over $\mathbb{Z}_p$ classifying elliptic curves with level structure,
- $X = X_B$ a Shimura curve of level $V_1(N)$ defined over $\mathbb{Z}_p$ classifying false elliptic curves with level structure.

We will denote $\overline{X} = X \times k$ its special fiber, $C \subset X(\mathbb{Z}_p)$ the finite collection (possibly empty) of cusps. Let ω be the ample line bundle over X obtained from the pushforward of the differential bundle on the universal object. Let $(\mathcal{H}, \nabla)$ be the rank 2 vector bundle arising from the relative de Rham cohomology over X with logarithmic connection

$$\nabla : \mathcal{H} \to \mathcal{H} \otimes_{\mathbb{Z}_p} \Omega^1_X(\log C)$$

coming from the Gauss-Manin connection. In the quaternionic case we are fixing a choice of a non-trivial idempotent element of $M_2(\mathbb{Z}_p)$ and projecting the four dimensional vector bundle coming from the relative de Rham cohomology to obtain $\mathcal{H}_k$ and analogously with ω. See [**Kas04**]. Denote by $\mathcal{H}_k$ its k-th symmetric tensor power $\mathcal{H}_k = \mathrm{Sym}^k\mathcal{H}$.

We recall the standard properties of $\mathcal{H}_k$. The vector bundles ω and $\mathcal{H}$ fit in a short exact sequence

$$0 \to \omega \to \mathcal{H} \to \omega^{-1} \to 0 \tag{1}$$

inducing a $(k+1)$-step decreasing filtration on H_k

$$\mathcal{H}_k = \mathrm{Fil}^0\mathcal{H}_k \supseteq \mathrm{Fil}^1\mathcal{H}_k \supseteq \cdots \mathrm{Fil}^k\mathcal{H}_k \cong \omega^k \supseteq \mathrm{Fil}^{k+1}\mathcal{H}_k = 0. \tag{2}$$

The graded pieces of the filtration are isomorphic to

$$Gr_i\mathcal{H}_k = F^{k-i}\mathcal{H}_k / F^{k+1-i}\mathcal{H}_k \cong \omega^{k-2i}.$$

The connection obeys *Griffith's transversality*

$$\nabla \mathrm{Fil}^{k-i}\mathcal{H}_k \subseteq \mathrm{Fil}^{k-i-1}\mathcal{H}_k \otimes_{\mathcal{O}_X} \Omega^1_X(\log C)$$

and that these maps induce a *Kodaira-Spencer* type of isomorphism of $\mathcal{O}_X$-modules for $p > k+1$

$$\nabla : \mathrm{Gr}_i\mathcal{H}_k \cong \mathrm{Gr}_{i+1}\mathcal{H}_k \otimes_X \Omega^1_X(\log C). \tag{3}$$

Following the usual notation, we write

$$M_k(X) = \Gamma(X, \omega^k), \quad S_k(X) = \Gamma(X, \omega^{k-2} \otimes \Omega^1_X).$$

We denote the complex arising from the connection by

$$\mathcal{H}^\bullet_k := \left[0 \to \mathcal{H}_k \to \mathcal{H}_k \otimes_{\mathcal{O}_X} \Omega^1_X\right]$$

and denote its hypercohomology groups by H^i_{dR}

$$H^i_{dR}(X,\mathcal{H}_k) := \mathbb{H}^i(X,\mathcal{H}_k^\bullet).$$

Lemma 1. — *We have a short exact sequence*

$$0\to\omega_{-k}\xrightarrow{\theta^{k+1}}\omega_{k+2}\to\mathcal{H}_k\otimes_X\Omega^1_X(\log C)/\nabla_k(\mathcal{H}_k)\to 0 \tag{4}$$

where the right map is compatible with inclusions and an Eichler-Shimura *short exact sequence of* $\mathcal{O}_K$*-modules*

$$0\to M_{k+2}(X)\to H^1_{\mathrm{dR}}(X,\mathcal{H}_k)\to S_{k+2}(X)^\vee\to 0.$$

Proof. — Consider the filtration $(\mathrm{Fil}^p\mathcal{H}_k^\bullet)_p$ of the complex $\mathcal{H}_k^\bullet$

$$\mathrm{Fil}^p\mathcal{H}_k^\bullet := \left[0\to\mathrm{Fil}^p\mathcal{H}_k\to\mathrm{Fil}^{p-1}\mathcal{H}_k\otimes_{\mathcal{O}_X}\Omega^1_X(\log C)\right].$$

and take the associated spectral sequence $E_0^{p,q}=\mathrm{Fil}^p\mathcal{H}_k^{p+q}/\mathrm{Fil}^{p+1}\mathcal{H}_k^{p+q}$. The only terms that survive in page 1 are

$$E_1^{0,0}\cong\mathcal{H}_k/\mathrm{Fil}^1\mathcal{H}_k,\quad E_1^{k+1,-k}\cong\mathrm{Fil}^k\mathcal{H}_k\otimes_{\mathcal{X}}\Omega^1_{\mathcal{X}}(\log C).$$

The spectral sequence degenerates at page $k+2$ converging to $E_{k+2}^{k+1,-k}=H^1(\mathcal{H}_k^\bullet)$. This gives rise to the short exact sequence

$$0\to\mathcal{H}_k/\mathrm{Fil}^1\mathcal{H}_k\xrightarrow{\theta^{k+1}}\mathrm{Fil}^k\mathcal{H}_k\otimes_{\mathcal{X}}\Omega^1_{\mathcal{X}}(\log\mathcal{C})\to\mathcal{H}_k\otimes_{\mathcal{X}}\Omega^1_{\mathcal{X}}(\log C)/\nabla_k(\mathcal{H}_k)\to 0.$$

The second exact sequence is obtained applying the spectral sequence to the hypercohomology of the graded pieces obtaining

$$\begin{aligned}0\to H^0_{\mathrm{dR}}(X,\mathcal{H}_k^\bullet)\to H^0(X,\omega^{-k})\to H^0(X,\omega^k\otimes\Omega^1_X(\log C))\to H^1_{\mathrm{dR}}(X,\mathcal{H}_k^\bullet)\to\\ \to H^1(X,\omega^{-k})\to H^1(X,\omega^k\otimes\Omega^1_X(\log C)))\to\cdots\end{aligned}$$

giving the desired exact sequence using ampleness of ω and Serre duality. □

Fix an effective divisor $S=\sum_{i=1}^n\alpha_i$ supported on the open curve $X\setminus C$ and defined over $\mathbb{Z}_p$,

$$\alpha_i:\mathrm{Spec}(\mathcal{O}_K)\to X\setminus C$$

for K a finite field extension over $\mathbb{Q}_p$ with distinct close points on the special fiber $\overline{X}$. We denote by $\overline{S}$ their reduction to the special fiber $\overline{X}$ and $S_{\mathbb{Q}_p}$ their restriction to the generic fiber. We have that the curve $U=X_{\mathbb{Q}_p}\setminus S_{\mathbb{Q}_p}$ is affine.

As a direct consequence of the lemma, the de Rham cohomology of $\mathcal{H}_k$ can be computed by

$$H^1_{\mathrm{dR}}(U,C),\mathcal{H}_k)=M_{k+2}^{\mathrm{mero},S}/\theta^{k+1}(M_{-k,\lambda}^{\mathrm{mero},S})$$

where we have used the notation $M_{k+2}^{\mathrm{mero},S}=\Gamma(U,\omega^{k+2})$ and should be thought as *meromorphic modular forms with poles along* S.

3. Rigid cohomological description

Consider X^{rig} to be the rigid analytification of X and the reduction map

$$\mathrm{red}:X^{\mathrm{rig}}\to\overline{X}(\overline{\mathbb{F}}_p)$$

where we fixed $\overline{k}$ algebraic closure of k. We then consider the wide open obtained from X^{rig} removing closed balls in the residue disks of the points S. Since X is smooth, $\mathrm{red}^{-1}(\overline{\alpha})$ is conformal to an open ball B for every $\overline{\alpha}\in\overline{S}$. Fixing this parametrization and fixing $0<\lambda<1$, define

$$V_\lambda=X^{\mathrm{rig}}\setminus\bigcup_{\alpha\in S}B(\alpha,\lambda)$$

with $B(\alpha,\lambda)$ closed ball inside $\mathrm{red}^{-1}(\overline{\alpha})$ centered in α. After an application of rigid GAGA to the coherent sheaves, the aciclycity of wide open neighborhoods implies that the hypercohomology of the complex $\mathcal{H}_k^\bullet$ can be computed simply by

$$H^1_{\mathrm{dR}}(V_\lambda,\mathcal{H}_k^{\mathrm{rig}})\cong\Gamma(V_\lambda,\mathcal{H}_k^{rig}\otimes\Omega^1_{X^{rig}}(\log C))/\nabla_k(\Gamma(V_\lambda,\mathcal{H}_k^{rig}))$$

Borrowing once again the notation from the modular curve case, we will denote the sections of ω^{rig} over V_λ by

$$M^{\dagger,S}_{k,\lambda} = \Gamma(V_\lambda, (\omega^{\mathrm{rig}})^k)$$

and they should be think as λ-*overconvergent modular forms.* We will denote it simply as $M^{\dagger}_{k,\lambda}$ when the set of poles is clear from the context.

We can apply the acyclicity property of V_λ to the short exact sequence of Lemma 1.

$$M^{\dagger}_{k+2,\lambda}/\theta^{k+1}(M^{\dagger}_{-k,\lambda}) \cong \Gamma(V_\lambda, \mathcal{H}^{rig}_k \otimes \Omega^1_{X^{rig}}(\log C))/\nabla_k(\Gamma(V_\lambda, \mathcal{H}^{rig}_k)) \cong H^1_{\mathrm{dR}}(X\setminus S, \mathcal{H}_k). \tag{5}$$

Since every vector bundle with connection on a smooth formal model X of X^{rig} is overconvergent, the work of Baldassarri-Chiarellotto [**BC91**] establishes a comparison isomorphism

$$H^1_{\mathrm{dR}}((X_{\mathbb{Q}_p}, S_{\mathbb{Q}_p}), \mathcal{H}_k) \cong H^1_{\mathrm{dR}}(V_\lambda, \mathcal{H}^{\mathrm{rig}}_k).$$

The left-hand side in the previous isomorphism is the de Rham cohomology with logarithmic coefficients. In characteristic 0, by an analytic result of Deligne [**Del70**] and later extended to an algebraic version in [**AKR07**] we can relate it to the open curve case

$$H^1_{\mathrm{dR}}(X_{\mathbb{Q}_p}\setminus S_{\mathbb{Q}_p}, \mathcal{H}_k) \cong H^1_{\mathrm{dR}}(V_\lambda, \mathcal{H}^{\mathrm{rig}}_k).$$

In particular, we have the following commutative diagram where all arrows are isomorphisms

$$\begin{tikzcd} M^{\mathrm{mero},S}_{k+2}/\theta^{k+1}(M^{\mathrm{mero},S}_{-k,\lambda}) \arrow[r] \arrow[d] & H^1_{\mathrm{dR}}(X_{\mathbb{Q}_p}\setminus S_{\mathbb{Q}_p}, \mathcal{H}_k) \arrow[d] \\ M^{\dagger}_{k+2,\lambda}/\theta^{k+1}(M^{\dagger}_{-k,\lambda}) \arrow[r] & H^1_{\mathrm{dR}}(V_\lambda, \mathcal{H}^{\mathrm{rig}}_k). \end{tikzcd} \tag{6}$$

In order to study and relate the Frobenius action on global sections and on the fibers we will use the structure of *overconvergent* log F-*isocrystal* of $(\mathcal{H}_k, \nabla)$. The relative de Rham cohomology in the modular curve case has this property thanks to the classical work of Katz [**Kat73**] while the Shimura curve case has been studied by Kassaei in [**Kas04**].

We then apply the Frobenius equivariant residue sequence following Coleman [**Col94**] and Coleman-Iovita [**CI10**].

$$0 \to H^1_{\mathrm{dR}}(X^{rig}, \mathcal{H}^{\mathrm{rig}}_k) \to H^1_{\mathrm{dR}}(V_\lambda, \mathcal{H}^{\mathrm{rig}}_k) \xrightarrow{\mathrm{Res}} \bigoplus_{\alpha\in S_K} H^0_{\mathrm{dR}}(A_{\alpha,\lambda}, \mathcal{H}_k)[1] \to 0 \tag{7}$$

Where $A_{\alpha,\lambda}$ are rigid subspaces of V_λ conformal to the open annuli $\mathrm{res}^{-1}(\overline{\alpha})\setminus B(\alpha,\lambda)$ and [1] refers to the shift in the Frobenius action. The previous exact sequence gives in fact a rigid analogue of the Gysin sequence. Since every residue disk admits a basis of orizontal section for an overconvergent isocrystal as proved by Katz in [**Kat71**], we can rewrite the exact sequence as

$$0 \to H^1_{\mathrm{dR}}(X^{rig}, \mathcal{H}^{\mathrm{rig}}_k) \to H^1_{\mathrm{dR}}(V_\lambda, \mathcal{H}^{\mathrm{rig}}_k) \xrightarrow{\mathrm{Res}} \bigoplus_{\alpha\in S_{\mathbb{Q}_p}} \mathrm{Sym^k}(\alpha^*\mathcal{H})[1] \to 0. \tag{8}$$

In particular we have that the cohomology groups have the following dimensions

- $\dim_{\mathbb{Q}_p} H^1_{\mathrm{dR}}(X^{rig}, \mathcal{H}^{\mathrm{rig}}_k) = \dim_{\mathbb{Q}_p} M_{k+2}(X_{\mathbb{Q}_p}) + \dim_{\mathbb{Q}_p} S_{k+2}(X_{\mathbb{Q}_p})$ by Lemma 1,
- $\dim_{\mathbb{Q}_p} \mathrm{Sym^k}(\alpha^*\mathcal{H}) = k+1$ for every $\alpha \in S_{\mathbb{Q}_p}$,
- $\dim_{\mathbb{Q}_p} H^1_{\mathrm{dR}}(V_\lambda, \mathcal{H}^{\mathrm{rig}}_k) = \dim_{\mathbb{Q}_p} H^1_{\mathrm{dR}}(X^{rig}, \mathcal{H}^{\mathrm{rig}}_k) + |S_{\mathbb{Q}_p}|(k+1)$.

In the elliptic setting we have

$$\mathrm{Sym^k}(\alpha^*\mathcal{H}) \cong \mathrm{Sym^k}(H^1_{\mathrm{dR}}(\mathcal{E}_\alpha))$$

for $\mathcal{E}_\alpha$ elliptic curve corresponding to the point α. In the quaternionic setting we have

$$\mathrm{Sym^k}(\alpha^*\mathcal{H}) \cong \mathrm{Sym^k}(e_1.H^1_{\mathrm{dR}}(\mathcal{A}_\alpha))$$

for $\mathcal{A}_\alpha$ abelian surface corresponding to the point α and e_1 a fixed choice of idempotent in $M_2(\mathbb{Z}_p)$.

4. Integral Frobenius structure

We now consider the log-crystalline cohomology group $H^1_{\log-\mathrm{crys}}(((\overline{X},\overline{S})/\mathbb{Z}_p),\mathcal{H}_k)$. Since $\mathcal{H}_k$ has the structure of a log F-crystal as studied by Ogus [**Ogu94**], we obtain a $\mathbb{Z}_p$ module with a Frobenius action. By the classical work of Kato [**Kat89**] we have a comparison theorem with de Rham cohomology

$$H^1_{\log-\mathrm{crys}}(((\overline{X},\overline{S})/\mathbb{Z}_p),\mathcal{H}_k)\otimes\mathbb{Q}_p\cong H^1_{\mathrm{dR}}((X_{\mathbb{Q}_p},S_{\mathbb{Q}_p}),\mathcal{H}_k)$$

and a comparison with rigid cohomology

$$H^1_{\log-\mathrm{crys}}(((\overline{X},\overline{S})/\mathbb{Z}_p),\mathcal{H}_k)\otimes\mathbb{Q}_p\cong H^1_{\mathrm{dR}}(V_\lambda,\mathcal{H}_k^{\mathrm{rig}})$$

compatible with the action of Frobenius. We then find a Frobenius stable $\mathbb{Z}_p$-lattice Λ in the rigid cohomology

$$\Lambda:=\mathrm{Im}(H^1_{\mathrm{crys}}((\overline{X},\overline{C}\cup\overline{S})/\mathbb{Z}_p,\mathcal{H}_k)\to H^1_{\mathrm{crys}}((\overline{X},\overline{C}\cup\overline{S})/\mathbb{Z}_p,\mathcal{H}_k)\otimes_{\mathbb{Z}_p}\mathbb{Q}_p\cong H^1_{\mathrm{dR}}(V_\lambda,\mathcal{H}_k^{\mathrm{rig}})).$$

By equation (5) we can see Λ contained in

$$\Lambda\hookrightarrow M^\dagger_{k+2,\lambda}/\theta^{k+1}(M^\dagger_{-k,\lambda}).$$

4.1. Explicit Frobenius lift. — Consider now the classical case of S consisting of the set supersingular points and $\lambda<1/p+1$. As explained by [**Kat73**] and [**Kas04**], there exist a section of the projection map $X(p)\to X$ induced by the *canonical subgroup*. This morphism gives a lift of Frobenius V on the modular curve X given by

$$V:V_\lambda\to V_{\lambda^p} \tag{9}$$

acting at the level of the moduli space sending the elliptic curve or abelian surface to the corresponding quotient by the canonical subgroup. In the case of the elliptic modular curve, the V operator acts on q-expansion by

$$(Vf)(q^{1/N})=p^kf(q^{p/N}).$$

The morphism V is flat and we can then define a U_p operator as the trace of Frobenius acting on the de Rham cohomology $H^1_{dR}(V_{\lambda^p},\mathcal{H}_k)$. Coleman proved the following theorem in the case of the elliptic modular curve.

Theorem 1 **([Col96] Theorem 5.4).** — *The space $H^1_{\mathrm{dR}}(V_\lambda,\mathcal{H}_k^{\mathrm{rig}})$ is finite dimensional and the operators V and U_p acting on the modular curve by correspondence induce endomorphisms* Frob *and* Ver *of $H^1_{\mathrm{dR}}(V_\lambda,\mathcal{H}_k^{\mathrm{rig}})$ such that*

$$\mathrm{Frob}\circ\mathrm{Ver}=\mathrm{Ver}\circ\mathrm{Frob}=p^{k+1}.$$

As further explained, the action of Verschibung on the cohomology group $H^1_{\mathrm{dR}}(V_{\lambda^p},\mathcal{H}_k^{\mathrm{rig}})$ is given by

$$\mathrm{Ver}:[f]\to[U_pf]$$

for $f\in M^\dagger_{k+2,\lambda}$ under the isomorphism (5)

$$M^\dagger_{k+2,\lambda}/\theta^{k+1}(M^\dagger_{-k,\lambda})\cong H^1_{\mathrm{dR}}(V_{\lambda,\mathcal{H}_k^{\mathrm{rig}})}.$$

This relation extends formally from the results of Kassaei for the Shimura curve over $\mathbb{Q}$.

Enlarging the set S and removing disks from the ordinary locus, we can restrict the action of Frobenius and U_p to a smaller domain. The key easy lemma for Frobenius neighborhoods is the following

Lemma 2. — *Let B the residue disk of an ordinary point α defined over $\mathbb{F}_p$, let t be a Serre-Tate parameter. Then $B\cong\mathrm{Spf}(\mathbb{Z}_p[[t]])_\eta$ and the Frobenius lift V restricted to B is a strict contraction.*

We can then find a small enough λ for which the Frobenius lift V acts as in (9). Combining the results of this section we obtain that there exists $f_1,\dots,f_r\in M^\dagger_{k+2,\lambda}$ λ-overconvergent modular forms with $r=\dim_{\mathbb{Q}_p}H^1_{\mathrm{dR}}(V_{\lambda^p},\mathcal{H}_k^{\mathrm{rig}})$ such that

$$\Lambda=\langle[f_i]:i=1,\dots,r\rangle_{\mathbb{Z}_p}$$

On which the V operator acts as an endomorphism and for which we have the relation $V \circ U_p = U_p \circ V = p^{k+1}$. Under the chain of isomorphisms (6) we can pick $f_1, \dots, f_r$ to be meromorphic modular forms over $\mathbb{Q}_p$ with poles along S.

We conlclude the section with a remark on known results about the slope of p-newforms.

Remark 1. — *Consider once again the case of S consisting of all the supersingular points of X. Then* [**Col96**] *in the case of elliptic modular curves identified the parabolic part of $H^1_{\mathrm{dR}}(V_{p/p+1}, \mathcal{H}_k)$ with classical modular forms of level $\Gamma_1(N) \cap \Gamma_0(p)$. This has been later generalized in* [**Kas09**] *including the case of quaternionic modular forms. Theorem 1 allows us to relate the slopes of eigenvalues of the Frobenius with the slopes of eigenvalues of U_p operator. In the elliptic case we obtain that the Frobenius eigenvalues are $\pm p^{\frac{k+1}{2}}$ on the space*

$$\bigoplus_{\alpha \in S_{\mathbb{Q}_p}} \mathrm{Sym}^{\mathrm{k}}(\alpha^* \mathcal{H}) \cong \bigoplus_{\alpha \in S_{\mathbb{Q}_p}} \mathrm{Sym}^k H^1_{\mathrm{dR}}(\mathcal{E}_\alpha)$$

where $\mathcal{E}_\alpha$ is the elliptic curve corresponding to the point α. From the exact sequence (8) we deduce that the newspace of the classical modular forms of level $\Gamma_1(N) \cap \Gamma_0(p)$ have slope $\frac{k-1}{2}$ due to the 1-shift.

5. Arithmetic applications

5.1. Congruences on q-expansion. — In this section we restrict to the case $X = X_1(N)$ elliptic modular curve and we consider the expansion at the cusp $[\infty]$. We summarize here the classical geometric description of q-expansion, referring to [**Kat73**] for further details. Consider the infinitesimal neighborhood $\mathrm{Spf}(\mathbb{Z}_p[[q]])$ of the cusp $[\infty]$ of the formal model $\mathfrak{X}$. The fiber product with the universal elliptic curve gives rise to the Tate curve

$$\begin{array}{ccc} \mathrm{Tate}(q^{1/N}) & \longrightarrow & \mathcal{E} \\ \big\downarrow & & \big\downarrow \\ \mathrm{Spf}(\mathbb{Z}_p[[q^{1/N}]]) & \xrightarrow{\iota} & \mathfrak{X}. \end{array}$$

The global sections of the line bundle are given by

$$\Gamma(\mathrm{Spf}(\mathbb{Z}_p[[q^{1/N}]]), \iota^*\omega) = \mathbb{Z}_p[[q^{1/N}]]\omega_{\mathrm{can}}$$

where ω_{can} is the canonical differential dt/t on the Tate curve $\mathbb{G}_m/q^{\mathbb{Z}}$. The global sections of the sheaf of differential 1-forms on X with logarithmic poles at the cusps are given by

$$\Gamma(\mathrm{Spf}(\mathbb{Z}_p[[q^{1/N}]]), \iota^*\Omega^1_{\mathcal{X}}(\log \mathcal{C})) = \mathbb{Z}_p[[q^{1/N}]]\frac{dq}{q}.$$

The Kodaira-Spencer isomorphism (3) gives an identification

$$KS(\omega^2_{\mathrm{can}}) = \frac{dq}{q}$$

Given a modular form $f \in \Gamma(U, \omega^k \otimes \Omega^1_X(\log C))$ with $U \subset X$ we can then define its q-expansion as

$$\iota^* f = f(q^{1/N})\omega^k_{\mathrm{can}} \otimes \frac{dq}{q}, \quad f(q^{1/N}) = \sum_{n=0}^{\infty} a_n(f) q^{\frac{n}{N}}.$$

The pull back of the relative de Rham cohomology at infinity is given by

$$\Gamma(\mathrm{Spf}(\mathbb{Z}_p[[q^{1/N}]]), \iota^*\mathcal{H}) = \mathbb{Z}_p[[q^{1/N}]])\omega_{\mathrm{can}} \oplus \mathbb{Z}_p[[q^{1/N}]])\eta_{\mathrm{can}}$$

where

$$\nabla(\omega_{\mathrm{can}}) =: \eta_{\mathrm{can}} \otimes \frac{dq}{q} \in \iota^*(\mathcal{H} \otimes \Omega^1_X), \qquad \nabla(\eta_{\mathrm{can}}) = 0.$$

In general we will have,

$$\Gamma(\mathrm{Spf}(\mathbb{Z}_p[[q^{1/N}]]), \iota^*\mathcal{H}_k) = \bigoplus_{a+b=k} \mathbb{Z}_p[[q^{1/N}]])\omega^a_{\mathrm{can}}\eta^b_{\mathrm{can}}$$

By the acyclicity of $\mathrm{Spf}(\mathbb{Z}_p[[q^{1/N}]])$, we have that

$$H^1_{\mathrm{dR}}(\mathrm{Spf}(\mathbb{Z}_p[[q^{1/N}]]), \iota^*\mathcal{H}_k) = \Gamma(\mathrm{Spf}(\mathbb{Z}_p[[q^{1/N}]]), \iota^*(\mathcal{H}_k \otimes_{\mathcal{X}} \Omega^1_{\mathcal{X}}))/\nabla(\Gamma(\mathrm{Spf}(\mathbb{Z}_p[[q^{1/N}]]), \iota^*(\mathcal{H}_k))).$$

Using the exact sequence in Lemma 1 we then obtain

$$\begin{aligned} H^1_{\mathrm{dR}}(\mathrm{Spf}(\mathbb{Z}_p[[q^{1/N}]]), \iota^*\mathcal{H}_k) &\cong \Gamma(\mathrm{Spf}(\mathbb{Z}_p[[q^{1/N}]]), \iota^*\omega^{k+2}/\theta^{k+1}(\Gamma(\mathrm{Spf}(\mathbb{Z}_p[[q^{1/N}]]), \iota^*\omega^{-k}))) \\ &\cong \mathbb{Z}_p[[q^{1/N}]]/\theta^{k+1}(\mathbb{Z}_p[[q^{1/N}]]). \end{aligned}$$

Combining the constructions we can the form a *q-expansion map* on the cohomology classes

Lemma 3. — *The morphism $\iota : \mathrm{Spf}(\mathbb{Z}_p[[q^{1/N}]]) \to \mathfrak{X}$ between formal schemes induces the following morphism*

$$\begin{aligned} \Lambda &\to \mathbb{Z}_p[[q^{1/N}]]/\theta^{k+1}(\mathbb{Z}_p[[q^{1/N}]]) \\ [f] &\mapsto [f(q)]. \end{aligned}$$

Proof. — By an integral comparison with the de Rham cohomology [**Kat89**] (Theorem 6.4) we can compute the crystalline cohomology by

$$H^1_{\mathrm{crys}}((\overline{X}, \overline{C} \cup \overline{S})/\mathbb{Z}_p, \mathcal{H}_k) \cong H^1_{\mathrm{dR}}((\mathfrak{X}, S), \mathcal{H}_k)$$

The map is then induced by the functoriality along the morphism ι

$$H^1_{\mathrm{dR}}((\mathfrak{X}, S), \mathcal{H}_k) \to H^1_{\mathrm{dR}}(\mathbb{Z}_p[[q^{1/N}]]), \iota^*\mathcal{H}_k).$$

and by the previous remark we have

$$H^1_{\mathrm{dR}}(\mathbb{Z}_p[[q^{1/N}]]), \iota^*\mathcal{H}_k) \cong \mathbb{Z}_p[[q^{1/N}]]/\theta^{k+1}(\mathbb{Z}_p[[q^{1/N}]]).$$

$\square$

Theorem 2. — *Let X be an elliptic modular curve of level N defined over $\mathbb{Z}_p$, with $N > 3$, $p > 3$ and $(N, p) = 1$. Fix α be a $\mathbb{Z}_p$ point on the modular curve. Let k be an integer with $p > k + 1$, let μ be an eigenvalue for the Frobenius action on $\mathrm{Sym}^k H^1_{\mathrm{dR}}(\alpha^*\mathcal{E})$ with $\mathcal{E} \to X$ universal elliptic curve. Then there exists a modular form $f \in M^{\mathrm{mero},\{\alpha\}}_{k+2} \otimes K$ for $K/\mathbb{Q}_p$ finite field extension and for which $[f] \in \Lambda \otimes_{\mathbb{Z}_p} \mathcal{O}_k$ is an eigenvector for the action of* Frob *with eigenvalue $p\mu$. Furthermore, for every $n, l > 1$*

$$a_{np^{l+1}}(f) \equiv \frac{p^k}{\mu} a_{np^l}(f) \mod p^{l(k+1)}.$$

Proof. — Consider the Gysin sequence obtained by removing α and the one enlarge by removing all the supersingular points SS. Let S be the set $SS \cup \{\alpha\}$ consider the exact sequences of $\mathbb{Q}_p$ vector spaces with Frobenius action

$$\begin{tikzcd} 0 \arrow[r] & H^1_{\mathrm{dR}}(X^{\mathrm{rig}}, \mathcal{H}^{\mathrm{rig}}_k) \arrow[r] & M^{\dagger,S}_{k+2,\lambda}/\theta^{k+1}(M^{\dagger,S}_{-k,\lambda}) \arrow[r] & \bigoplus_{\alpha \in SS_{p,\mathbb{Q}_p}} \mathrm{Sym}^k H^1_{\mathrm{dR}}(\mathcal{E}_\alpha) \oplus \mathrm{Sym}^k H^1_{\mathrm{dR}}(\alpha^*\mathcal{E})[1] \arrow[r] & 0 \\ 0 \arrow[r] & H^1_{\mathrm{dR}}(X^{\mathrm{rig}}, \mathcal{H}^{\mathrm{rig}}_k) \arrow[r] \arrow[u, "="] & M^{\dagger,\{\alpha\}}_{k+2,\lambda}/\theta^{k+1}(M^{\dagger,\{\alpha\}}_{-k,\lambda}) \arrow[r] \arrow[u] & \mathrm{Sym}^k H^1_{\mathrm{dR}}(\alpha^*\mathcal{E})[1] \arrow[r] \arrow[u] & 0 \end{tikzcd}$$

We can then consider the eigenclass $[f]$ representing the eigenvector attached to $p\mu$ in $M^{\dagger,\{\alpha\}}_{k+2,\lambda}/\theta^{k+1}(M^{\dagger,\{\alpha\}}_{-k,\lambda})$. The residue map shift the Frobenius by one multiplying the eigenvalues by p. Under the comparison isomorphism with the de Rham cohomology (6), we can pick $f \in M^{\mathrm{mero},\{\alpha\}}_{k+2}$. Using Theorem 1, we can explictly compute the action of Frobenius and of U_p on the q-expansion. The action of U_p, trace of the Frobenius lift V on the q-expansion of weight $k + 2$ modular forms, is classically given by

$$f(q^{1/N}) = \sum_{n \geq 0} a_n(f) q^{n/N} \mapsto (U_p f)(q^{1/N}) = \sum_{n \geq 0} a_{pn}(f) q^{n/N}.$$

Taking $[f]$ to be an eigenclass in Λ corresponding to an eigenvalue $p\mu$ for the action of Frobenius we then have $[f]$ is an eigenclass of the U_p operator with eigenvalue $p^{k+1}/p\mu$ by Theorem 1

$$\left(\frac{p^{k+1}}{p\mu} f - U_p f\right)(q^{1/N}) \in \theta^{k+1}\mathbb{Z}_p[[q^{1/N}]]$$

from which we deduce for $l > 1$

$$\frac{p^k}{\mu} a_{np^l}(f) - a_{np^{l+1}}(f) \equiv 0 \mod p^{l(k+1)}.$$

□

5.2. An explicit example - Zhang's congruences. — In [**Zha25**], Zhang provided numerous numerical observation about meromorphic modular forms and their relation with the Frobenius eigenvalues of the poles. In this short section we want to show how the previous constructions and results provide a framework to understand his congruences.

We will work directly with the coarse moduli space $X(1) \cong \mathbb{P}^1_j$. The descent of the results from level N to level 1 can be carried over by classical techniques. We consider Example 4.6 of *loc. cit.*. Fix $k = 2$ and $p > k + 1 = 3$, $C/\mathbb{Q}$ an elliptic curve given by the equation

$$y^2 + xy = x^3 - x^2 - 2x - 1.$$

We have C is a CM elliptic curve with respect to the order $\mathbb{Z}[\frac{1+\sqrt{-7}}{2}]$ of discriminant -7 and $j(C) = -3375$. Consider $S = \{j(C)\} \cup SS_p$ where SS_p consists in a set of lift of supersingular points on $X(1)$. The Gysin sequence (8) gives us the following commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & H^1_{\mathrm{dR}}(X^{\mathrm{rig}}, \mathcal{H}_2^{\mathrm{rig}}) \arrow[r] \arrow[d, "="] & M^{\dagger, SS_p}_{4,p^{1/p+1}}/\theta^3(M^{\dagger, SS_p}_{-2,p^{1/p+1}}) \arrow[r] \arrow[d] & \bigoplus_{\alpha \in SS_{p,\mathbb{Q}_p}} \mathrm{Sym}^2 H^1_{\mathrm{dR}}(\mathcal{E}_\alpha)[1] \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & H^1_{\mathrm{dR}}(X^{\mathrm{rig}}, \mathcal{H}_2^{\mathrm{rig}}) \arrow[r] & M^{\dagger, S}_{4,\lambda}/\theta^3(M^{\dagger, S}_{-k,\lambda}) \arrow[r] & \bigoplus_{\alpha \in SS_{p,\mathbb{Q}_p}} \mathrm{Sym}^2 H^1_{\mathrm{dR}}(\mathcal{E}_\alpha) \oplus \mathrm{Sym}^2 H^1_{\mathrm{dR}}(C)[1] \arrow[r] & 0 \\
0 \arrow[r] & H^1_{\mathrm{dR}}(X^{\mathrm{rig}}, \mathcal{H}_2^{\mathrm{rig}}) \arrow[r] \arrow[u, "="] & M^{\dagger, \{j(C)\}}_{4,p^{1/p+1}}/\theta^3(M^{\dagger, \{j(C)\}}_{-2,p^{1/p+1}}) \arrow[r] \arrow[u] & \mathrm{Sym}^2 H^1_{\mathrm{dR}}(C)[1] \arrow[r] \arrow[u] & 0
\end{tikzcd}$$

The radius λ can be chosen accordingly to Lemma 2 to have an explicit lift of Frobenius corresponding with the quotient of the canonical subgroup. We have an isomorphism of $\mathbb{Q}_p$-vector spaces

$$\bigoplus_{\alpha \in SS_{p,\mathbb{Q}_p}} \mathrm{Sym}^2 H^1_{\mathrm{dR}}(\mathcal{E}_\alpha) \cong S_4(\Gamma_0(p))^{p-\mathrm{new}}$$

this identification is at the core of the proof of classicality of Coleman.

The bottom row can be identified with the equivalent Gysin exact sequence in de Rham cohomology, in particular we have the isomorphism

$$M_4^{\mathrm{mero}, \{j(C)\}}/\theta^3 \left(M_{-2,\lambda}^{\mathrm{mero}, \{j(C)\}} \right) \cong M^{\dagger, \{j(C)\}}_{4,p^{1/p+1}}/\theta^{k+1} \left(M^{\dagger, \{j(C)\}}_{-2,p^{1/p+1}} \right)$$

The three dimensional space given by $\mathrm{Sym}^2 H^1_{\mathrm{dR}}(C)$ can be represented in the cohomology group by meromorphic modular forms with poles along $j(C) = -3375$. Zhang identifies the following basis

$$\begin{aligned}
f_1 &= \frac{E_4}{j + 3375}, \\
f_2 &= 19 \cdot \frac{E_4}{j + 3375} - 91125 \cdot \frac{E_4}{(j + 3375)^2}, \\
f_3 &= 1399 \cdot \frac{E_4}{j + 3375} - 19008675 \cdot \frac{E_4}{(j + 3375)^2} + 54251268750 \cdot \frac{E_4}{(j + 3375)^3}.
\end{aligned}$$

Then for every ordinary prime $p > 3$ we have

$$\begin{aligned}
a_{np^{l+1}}(f_1) &\equiv u_p(C)^2 a_{np^l}(f_1) \mod p^{3l}, \\
a_{np^{l+1}}(f_2) &\equiv p a_{np^l}(f_1) \mod p^{3l}, \\
a_{np^{l+1}}(f_3) &\equiv p^2 u_p(C)^{-2} a_{np^l}(f_1) \mod p^{3l}
\end{aligned}$$

where $u_p(C)$ is the p-adic root of the polynomial $X^2 - a_p(C)X + p$. According to Theorem 2, the meromorphic modular forms f_1, f_2, f_3 are the representatives of the classes in $H^1_{\mathrm{dR}}(V_\lambda, \mathcal{H}_k^{\mathrm{rig}})$ corresponding to the Frobenius eigenspace isomorphic to $\mathrm{Sym}^2 H^1_{\mathrm{dR}}(C)$.

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